

The Guardian: A Modular Real-Time Framework for Proactive Human-Layer Cybersecurity and Contextual Risk Inference on Social Media

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Abstract—The increasing use of social media platforms has introduced significant human-layer cybersecurity risks, including misinformation, hate speech, phishing attacks, and unintended disclosure of sensitive personal information. Existing solutions typically address these challenges in isolation and are not designed to operate in real-time user interaction environments.

This paper presents *The Guardian*, a modular real-time framework for proactive human-layer cybersecurity on social media. The system integrates multiple detection modules, including misinformation detection, hate speech classification, scam and phishing detection, and personally identifiable information (PII) with contextual operational security risk inference, within a unified architecture. The framework operates as a browser-based extension connected to a backend analysis service, enabling real-time content analysis and user feedback during content creation.

The proposed approach combines machine learning models with rule-based techniques to detect both explicit and contextual risks. The system provides structured outputs, including risk classification, explanations, and safer rewrite suggestions, to support informed user decision-making.

System-level evaluation using realistic social media scenarios demonstrates an overall detection accuracy of 72%, with strong performance in identifying privacy and contextual risk signals. The results highlight the effectiveness of the modular and hybrid design while also revealing limitations in handling ambiguous cases.

The Guardian demonstrates the feasibility of integrating multiple human-layer cybersecurity capabilities into a single user-facing system, providing a foundation for future research in proactive and user-centric online safety solutions.

Index Terms—Human-layer cybersecurity, social media security, misinformation detection, hate speech detection, phishing detection, PII detection, operational security (OpSec), contextual risk inference, real-time systems, browser extension

I. INTRODUCTION

The rapid growth of social media platforms has transformed how individuals communicate, share information, and express opinions. However, this increased connectivity has also introduced new forms of cybersecurity risks that target users directly rather than technical systems. These human-layer threats include misinformation, hate speech, phishing scams, and unintended disclosure of sensitive personal information.

Unlike traditional cyber attacks that exploit system vulnerabilities, modern threats increasingly leverage human behavior, cognitive biases, and social interactions [1], [2]. Users are often exposed to deceptive content, manipulated narratives, and socially engineered attacks that are difficult to detect using conventional security mechanisms. At the same time, individuals frequently share personal information online without fully understanding the potential risks, leading to privacy exposure and exploitation.

Existing solutions typically address these problems in isolation. For example, separate systems have been developed for misinformation detection [3], hate speech classification [4], phishing detection [5], and privacy protection [6]. While these approaches have demonstrated effectiveness within their respective domains, they lack integration and are not designed to operate in real-time user interaction environments.

Furthermore, most existing tools focus solely on detection, providing limited support for user understanding or decision-making. Security warnings are often ignored or misunderstood by users, reducing their effectiveness [7], [8]. This highlights the need for systems that not only identify risks but also provide clear explanations and actionable guidance.

To address these challenges, this paper proposes *The Guardian*, a modular real-time framework for proactive human-layer cybersecurity on social media platforms. The system integrates multiple detection modules, including misinformation, hate speech, scam/phishing detection, and PII with contextual OpSec risk inference, into a unified architecture.

The Guardian operates as a browser-based extension connected to a backend analysis service. It analyzes user-generated content in real time and provides structured feedback, including risk classification, explanations, and safer alternatives. By combining multiple detection capabilities with user-centric feedback mechanisms, the system aims to assist users in making informed decisions before sharing content.

The main contributions of this work are as follows:

- A modular architecture that integrates multiple human-layer cybersecurity detection tasks into a unified real-time framework
- A hybrid detection approach combining machine learning and rule-based techniques across different risk categories
- A contextual risk inference mechanism for identifying implicit privacy and OpSec-related risks
- A user-centric feedback system that provides explanations and rewrite suggestions to support informed decision-making
- A comprehensive system-level evaluation demonstrating the effectiveness and limitations of the proposed approach

Unlike existing systems that focus on single-task detection, The Guardian introduces a unified, real-time, user-facing framework that combines multi-domain threat detection with contextual risk inference and actionable user guidance. This integration enables proactive intervention at the point of content creation, which remains underexplored in current human-layer cybersecurity research.

The remainder of this paper is organized as follows: Section II reviews related work, Section III describes the system architecture, Sections IV and V present module design and implementation, Section VI outlines the evaluation methodology, Section VII presents experimental results, Section VIII discusses the findings, and Section IX concludes the paper.

II. RELATED WORK

A. Human-Layer Cybersecurity and Social Engineering

Traditional cybersecurity solutions primarily focus on system-level threats such as malware and network attacks. However, recent research highlights that human behavior remains a critical vulnerability in modern digital environments. Social engineering attacks exploit user trust, cognitive biases, and behavioral patterns to achieve malicious objectives [1], [2].

User interaction with security mechanisms is another key challenge. Studies have shown that users often ignore or misunderstand security warnings, reducing their effectiveness [7], [8]. This emphasizes the need for systems that not only detect risks but also communicate them effectively to users in a timely and understandable manner.

B. Misinformation Detection

Misinformation detection has been widely studied using machine learning and natural language processing techniques. Early approaches relied on linguistic and stylistic features, while recent methods leverage deep learning models such as transformers for improved performance.

Datasets such as LIAR [9], FakeNewsNet [10], and CoAID [11] have been widely used to train and evaluate models. Studies have shown that misinformation spreads faster and more broadly than factual information [12], highlighting the importance of automated detection systems.

Despite these advances, challenges remain in detecting subtle manipulation, context-dependent misinformation, and evolving narratives.

C. Hate Speech and Toxicity Detection

Hate speech detection is a complex task due to the subjective and context-dependent nature of offensive language. Research has demonstrated that distinguishing between hate speech and general offensive language is non-trivial [4].

Recent datasets such as HateXplain [13] incorporate explainability to improve model transparency, while other studies address bias and generalization issues in hate speech detection systems [14]. Additionally, implicit and nuanced forms of abusive language remain difficult to detect [15].

These challenges highlight the need for robust and context-aware detection systems.

D. Scam and Phishing Detection

Phishing and scam detection techniques traditionally rely on URL analysis, blacklists, and heuristic rules. However, modern attacks increasingly use sophisticated language and social engineering tactics to deceive users.

Machine learning-based approaches have been proposed to detect malicious URLs and phishing content [5]. Hybrid methods combining structural features and textual analysis have shown improved performance [2].

Nevertheless, the dynamic nature of phishing attacks and the reliance on user interaction continue to pose significant challenges.

E. Privacy, PII, and Contextual Risk Exposure

Privacy research has emphasized that risk is not solely determined by explicit data disclosure but also by contextual information. The concept of contextual integrity highlights that information sharing becomes risky when it violates expected norms of context [6].

Studies in privacy and human behavior demonstrate that users often unintentionally disclose sensitive information online [16]. Furthermore, research on de-anonymization and inference attacks shows that seemingly harmless data can be combined to reveal sensitive information [17], [18].

Recent datasets and benchmarks such as PII-Bench [19] and PANORAMA [20] explore modern approaches to evaluating privacy risks and PII exposure in user-generated content.

F. Summary and Research Gap

Existing research has made significant progress in individual areas such as misinformation detection, hate speech classification, phishing detection, and privacy protection. However, these approaches are typically developed and deployed in isolation.

There is a lack of integrated systems that combine multiple detection capabilities into a unified, real-time framework that operates directly within user interaction workflows. Additionally, most existing solutions focus on detection alone, with limited emphasis on user guidance, explanation, and proactive risk mitigation.

To address these gaps, The Guardian framework proposes a modular, real-time system that integrates multiple detection modules and provides user-centric feedback, enabling proactive human-layer cybersecurity on social media platforms.

III. SYSTEM OVERVIEW

A. Design Goals

The Guardian framework is designed to provide real-time, user-facing protection against human-layer cybersecurity risks on social media platforms. The primary objectives of the system are:

- Detect multiple categories of online threats, including misinformation, hate speech, phishing, and privacy risks
- Provide real-time feedback during content creation
- Assist users in understanding and mitigating risks rather than blocking actions
- Maintain modularity for scalability and extensibility

B. Overall Architecture

The Guardian system follows a client-server architecture consisting of a browser extension and a backend analysis service. The browser extension captures user-generated content and communicates with the backend for analysis.

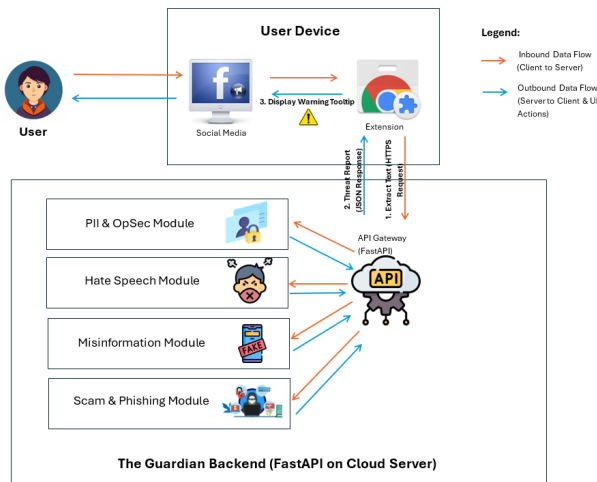


Fig. 1. Overall architecture of The Guardian framework showing browser extension, backend services, and modular analysis pipeline.

As illustrated in Fig. 1, the system processes user-generated content through a modular backend pipeline to detect multiple categories of risks. The architecture highlights the interaction between the browser extension, modular backend services, and the unified response generation pipeline, enabling real-time analysis and user-centric feedback delivery.

The backend processes input using multiple specialized modules and returns a structured response containing detected risks, explanations, and recommended actions.

C. Browser Extension Workflow

The browser extension operates as the primary interaction interface between the user and the system. It monitors text input fields on supported platforms and extracts content in real time.

The extracted content is sent to the backend using HTTP requests. The backend response is then displayed to the user through visual alerts, explanations, and suggested rewrites.

D. Backend and Module Integration

The backend is implemented using a modular FastAPI-based architecture. Each module independently analyzes the input text and produces task-specific outputs.

The outputs are then aggregated into a unified structure, ensuring consistency across different detection types.

E. Unified Response Contract

The backend returns a standardized JSON response that includes:

- Risk level classification
- Detected risk categories
- Highlighted content segments
- Explanations for identified risks
- Suggested safer alternatives

This unified contract ensures seamless integration with the browser extension and consistent user experience.

IV. MODULE DESIGN

The Guardian system consists of four specialized modules, each designed to address a specific category of human-layer cybersecurity risk.

A. PII and OpSec Risk Detection Module

This module focuses on detecting both explicit personal information and contextual exposure risks.

Explicit PII Detection: Structured and unstructured personal data such as phone numbers, email addresses, and names are detected using a hybrid approach combining regular expressions and Named Entity Recognition (NER).

Contextual OpSec Detection: Contextual signals such as travel intent, routine disclosure, real-time location sharing, and vulnerability expressions are identified using machine learning models and rule-based analysis.

The module combines these signals to infer potential risks from an attacker perspective.

The internal structure of this module is illustrated in Fig. 2.

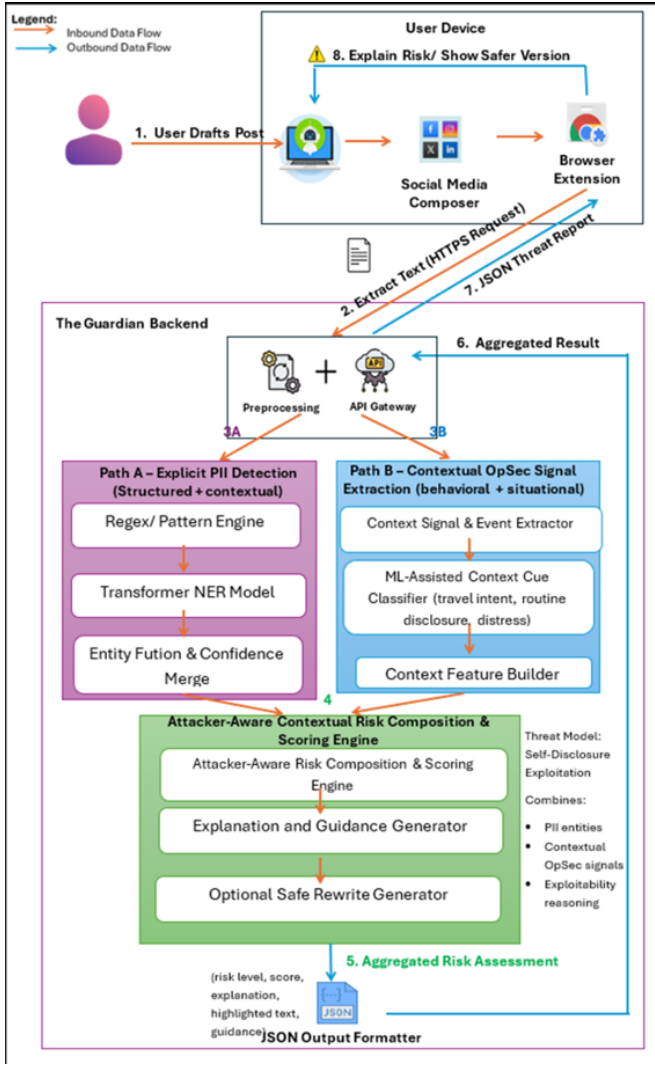


Fig. 2. Architecture of the PII and OpSec risk inference module showing explicit PII detection and contextual OpSec signal extraction paths.

B. Misinformation Detection Module

The misinformation module adopts a hybrid and user-centric design that extends beyond binary misinformation classification. Its core detection engine uses a fine-tuned DistilBERT-based transformer classifier to identify misleading or false content from contextual linguistic patterns such as exaggeration, emotional bias, manipulation, and unsupported claims.

To improve explainability, the module incorporates a dual evidence retrieval mechanism. First, an offline similarity-based retrieval layer uses TF-IDF matching to identify related claims from stored benchmark-style evidence. Second, an online evidence layer supports integration with external fact-checking sources to retrieve verified claims, supporting examples, source links, and credibility cues.

A key feature of this module is susceptibility-aware personalization. The system maintains a user profiling and susceptibility component that considers demographic and interaction-based factors such as age group, education level, and prior

feedback behavior. This produces a susceptibility score that influences the response strategy.

Based on the assessed risk and susceptibility level, the module generates adaptive chatbot-style explanations. Low-risk users may receive concise feedback, while higher-risk users receive more detailed explanations, simplified guidance, and stronger corrective suggestions. This design positions the misinformation module not only as a classifier, but also as an interactive educational intervention for improving user awareness and decision-making.

The architecture of the misinformation detection module is illustrated in Fig. 3.

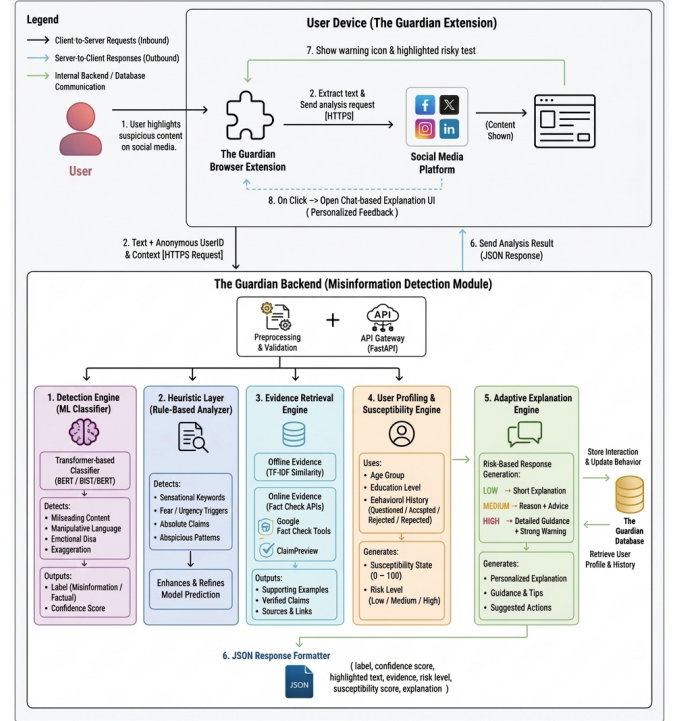


Fig. 3. Architecture of the misinformation detection module illustrating classification and verification components.

C. Hate Speech Detection Module

The hate speech module classifies text into categories such as offensive, abusive, or harmful content. It is trained to detect both explicit toxic language and contextual variations. The module contributes to the system by identifying content that may violate community guidelines or cause harm.

This module goes beyond simple keyword detection by understanding the context in which words are used. It can identify indirect or subtle forms of harmful language, such as sarcasm or coded expressions. By analyzing patterns in text, it improves the accuracy of detecting inappropriate content. The system helps reduce false positives by considering the overall meaning of sentences. Once harmful content is identified, appropriate actions such as flagging or removal can be taken. This ensures a safer and more respectful environment for users.

The architecture of the hate speech detection module is illustrated in Fig. 4.

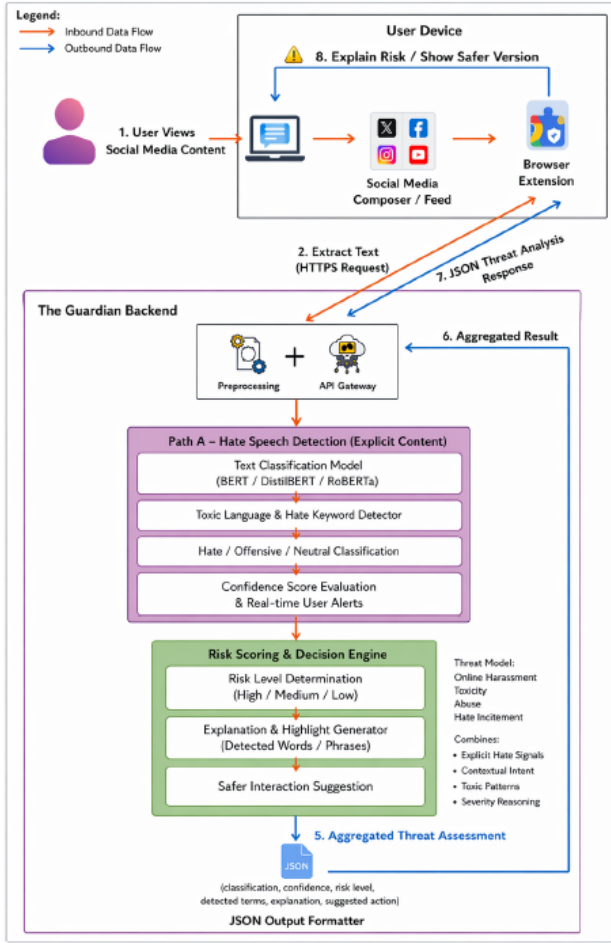


Fig. 4. Architecture of the hate speech detection module illustrating classification pipeline.

D. Scam and Phishing Detection Module

This module combines URL-based analysis with text-based classification to detect phishing attempts and scams.

It identifies suspicious links, urgency-driven language, impersonation patterns, and common social engineering tactics.

Blacklist and reputation-based services such as PhishTank [25] and Google Safe Browsing [26] represent commonly used external resources for phishing and malicious URL identification. The Guardian scam/phishing module is designed to remain compatible with such external threat-intelligence sources while retaining its own text-based and heuristic detection pipeline.

The hybrid approach improves detection reliability in real-world scenarios.

The structure of the scam and phishing detection module is shown in Fig. 5.

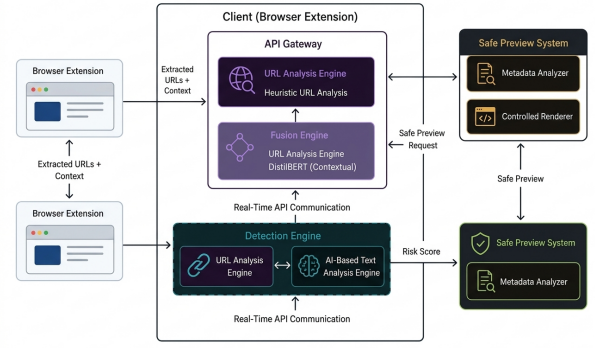


Fig. 5. Architecture of the scam and phishing detection module combining URL analysis and text classification.

V. IMPLEMENTATION

A. Frontend Extension

The browser extension is implemented using JavaScript and integrates directly with web pages to capture user input. It monitors text fields and triggers analysis when content is entered or modified.

The extension communicates with backend services through asynchronous HTTP requests and displays results using a user-friendly interface.

B. Backend Services

The backend is built using FastAPI, providing a scalable and modular architecture for handling requests from the extension.

Each detection module operates as an independent service or component, enabling parallel processing and efficient integration.

C. Model Deployment

Machine learning models used across modules are deployed as part of the backend services. Transformer-based models such as BERT [21], DistilBERT [22], and RoBERTa [23] are used for classification and contextual language understanding tasks. The Hugging Face Transformers library supports standardized deployment and experimentation with these model families [24].

Models are optimized for real-time inference to ensure low latency and responsiveness in the user interface.

D. Explanation and Guidance Generation

The system includes an explanation layer that translates model outputs into user-understandable feedback. This includes identifying risky content segments and explaining the associated risks.

Additionally, a rewrite generation component suggests safer alternatives that preserve the original intent while reducing exposure to potential threats.

This approach enhances usability by enabling users to take corrective actions rather than simply receiving warnings.

VI. EVALUATION METHODOLOGY

A. Module-Level Evaluation

Each Guardian module was evaluated independently using task-specific datasets and performance metrics appropriate to its detection objective. The misinformation, hate speech, and scam/phishing modules were assessed as supervised text classification tasks, while the PII and contextual OpSec module was evaluated through both component-level and system-level experiments.

For the PII and OpSec module, Path A focused on explicit PII detection using a hybrid approach that combines rule-based matching and named entity recognition. The PANORAMA dataset was used as a privacy-preserving base corpus because it provides a large-scale synthetic collection of realistic online text containing naturally embedded PII across content types such as social media posts, reviews, comments, and marketplace listings [20]. Since PANORAMA does not provide entity-level annotations, weak labeling was applied to construct training data for structured and contextual PII detection.

Path B focused on contextual OpSec risk detection. The OpSec dataset was iteratively hardened using realistic, ambiguous, adversarial, and multi-label cases to better reflect real-world disclosure scenarios. Model comparison included transformer-based classifiers such as BERT, DistilBERT, and RoBERTa, which were evaluated using exact match, micro F1, and macro F1.

B. System-Level Evaluation

System-level evaluation was conducted using the integrated Guardian extension and backend pipeline. A curated set of 25 realistic social media test cases was created to represent Safe, PII, OpSec, Combined, and Ambiguous scenarios. Each test case was passed through the deployed system, and the returned outputs were recorded for analysis.

For each input, the evaluation captured the predicted risk level, detected signals, explanation, advice, safe rewrite, and processing time. The actual risk classification was compared against the expected risk classification to determine correctness. Manual review was then used to assess the quality of explanations and rewrites, since these outputs require human judgment beyond automatic label matching.

C. Evaluation Metrics

The evaluation used both quantitative and qualitative metrics. Detection performance was assessed using accuracy and category-wise correctness. Component-level model experiments additionally used precision, recall, micro F1, macro F1, and exact match where appropriate.

Explanation quality was assessed using three levels: good, partial, and poor. A good explanation clearly identified the relevant risk and explained why it mattered. A partial explanation identified the general risk but lacked specificity or included partially irrelevant reasoning. A poor explanation was unclear, misleading, or not aligned with the detected risk.

Rewrite quality was evaluated using three dimensions: risk reduction, meaning preservation, and naturalness. A good

rewrite reduced risky disclosure while preserving the user's intended meaning and remaining natural. A partial rewrite preserved meaning but only partially reduced risk, or reduced risk while slightly weakening naturalness. A poor rewrite either failed to reduce risk, removed too much meaning, or produced an unnatural suggestion.

This mixed evaluation approach was selected because The Guardian is not only a detection system but also a user-facing decision-support system. Therefore, system effectiveness depends on both classification accuracy and the usefulness of the feedback presented to users.

VII. RESULTS AND EVALUATION

The Guardian framework was evaluated at both module level and system level to assess detection performance, contextual inference capability, and overall effectiveness in real-world social media scenarios.

A. Module-Level Results

Each module was evaluated independently using task-specific datasets and standard metrics. Table I summarizes the module-level performance.

TABLE I
MODULE-LEVEL EVALUATION SUMMARY

Module	Accuracy	F1-score
Misinformation Detection	98.85%	0.988
Hate Speech Detection	92.01%	0.783
Scam and Phishing Detection	98.63%	0.986
PII Detection (spaCy baseline)	–	0.866
OpSec Detection (Transformer models)	–	0.92–0.96

These results demonstrate that structured and pattern-based tasks achieve higher accuracy, while context-dependent tasks such as hate speech and OpSec detection remain more challenging.

The misinformation and scam/phishing modules achieved very high performance, with F1-scores of 0.988 and 0.986 respectively, indicating strong capability in detecting structured and pattern-based threats. The hate speech module achieved an accuracy of 92.01% with a macro F1-score of 0.783, reflecting the inherent difficulty of detecting context-dependent and implicit toxicity.

For PII detection, the spaCy-based NER model achieved an F1-score of 0.866, demonstrating strong performance in structured and semi-structured entity detection. Transformer-based models provided strong contextual modeling capabilities, while the spaCy-based NER baseline achieved the highest reported F1-score for structured PII detection in the evaluated setting, motivating the hybrid approach. The contextual OpSec detection models achieved macro F1-scores in the range of 0.92–0.96, indicating strong capability in identifying implicit privacy risks from contextual signals.

As shown in Fig. 6, the performance varies across modules depending on task complexity.

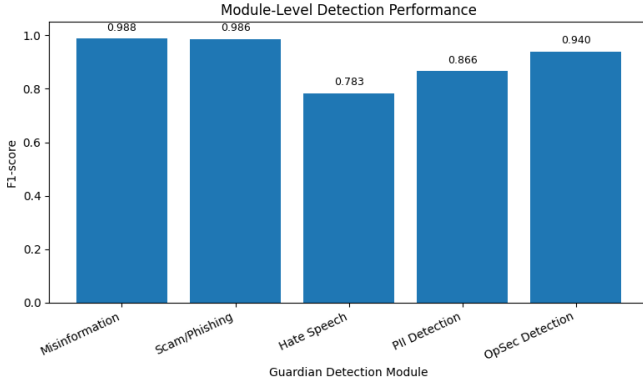


Fig. 6. Module-level F1-score comparison across Guardian components. The PII + OpSec module is decomposed into explicit PII detection (Path A) and contextual OpSec inference (Path B).

B. System-Level Evaluation

The integrated system was evaluated using a curated dataset of 25 realistic social media scenarios representing Safe, PII, OpSec, Combined, and Ambiguous cases.

The system achieved an overall accuracy of 72%, correctly classifying 18 out of 25 test cases.

Category-wise performance is as follows:

- Safe: 40%
- PII: 80%
- OpSec: 80%
- Combined: 100%
- Ambiguous: 80%

The results indicate strong performance in detecting explicit and contextual risks, while lower accuracy in safe scenarios reflects a conservative detection strategy that prioritizes minimizing potential privacy exposure over avoiding false positives.

C. Explanation and Rewrite Evaluation

In addition to detection performance, the system was evaluated on its ability to provide user-understandable explanations and actionable rewrite suggestions.

Explanation Quality:

- Good: 12 cases
- Partial: 12 cases
- Poor: 1 case

Most explanations were either clear or partially informative, demonstrating the system’s ability to communicate risk effectively.

Rewrite Quality:

- Good: 10 cases
- Partial: 5 cases
- Poor: 10 cases

Rewrite evaluation shows that while the system often preserves user intent and reduces risk exposure, performance varies in complex contextual scenarios.

D. Error Analysis

Error analysis reveals several recurring failure patterns:

- **Over-triggering in Safe Cases:** Benign content is sometimes classified as risky, particularly when location or temporal cues are present.
- **Ambiguity Handling Limitations:** Indirect or context-dependent expressions lead to misclassification or partial detection.
- **Partial Signal Coverage:** Some multi-signal scenarios are only partially detected, affecting overall risk assessment.
- **Vulnerability Over-interpretation:** Emotional expressions are occasionally misclassified as vulnerability signals.
- **Address Detection Inconsistency:** Informal or incomplete address patterns are not consistently identified.

Overall, the system demonstrates high sensitivity in detecting potential risks while maintaining moderate precision in distinguishing safe from ambiguous content.

VIII. DISCUSSION

The results demonstrate that The Guardian framework is effective in addressing a critical gap in existing cybersecurity solutions—namely, the lack of real-time, user-facing protection against human-layer risks on social media platforms.

A. Effectiveness of the Modular Architecture

The modular design of the system enables specialized detection of diverse threat categories, including misinformation, hate speech, phishing, and privacy-related risks. Each module contributes independently to the overall system while remaining interoperable through a unified backend architecture. This design not only improves maintainability and scalability but also allows for future extensions without disrupting existing components.

The strong performance observed across individual modules highlights the effectiveness of combining domain-specific detection strategies within a single integrated framework.

B. Hybrid Detection and Contextual Reasoning

A key contribution of this work lies in the integration of rule-based and machine learning-based techniques across modules. The results indicate that hybrid approaches are particularly effective in handling both structured signals (e.g., explicit identifiers, URLs) and unstructured contextual patterns (e.g., behavioral cues, linguistic signals).

In the PII and OpSec module, the combination of pattern-based detection and contextual inference enables the system to identify risks that extend beyond explicit data exposure. This is especially important in social media environments, where sensitive information is often revealed indirectly through context rather than direct disclosure.

C. System-Level Behavior and Trade-offs

The system-level evaluation reveals a deliberate trade-off between sensitivity and precision. While the framework achieves strong performance in detecting PII and contextual OpSec risks, its lower accuracy in safe scenarios reflects a tendency toward conservative classification.

This behavior is consistent with the system's design objective of prioritizing user safety. In human-layer cybersecurity, false negatives (i.e., missed risks) can have more severe consequences than false positives. Therefore, the system favors higher sensitivity, even at the cost of occasional over-triggering.

D. Multi-Signal Risk Inference

An important observation from the evaluation is the system's ability to handle multi-signal scenarios. Cases involving combinations of signals, such as temporal cues and personal identifiers, are more effectively classified compared to isolated signals. This suggests that modeling interactions between features improves overall risk assessment.

Unlike traditional single-signal detection systems, The Guardian framework incorporates contextual relationships between signals, enabling more realistic and attacker-relevant risk inference.

E. User-Centric Design and Explainability

Beyond detection accuracy, the system emphasizes usability through explanation and rewrite generation. The evaluation shows that the system can provide meaningful feedback that helps users understand why content is risky and how it can be improved.

This user-centric approach distinguishes The Guardian from conventional security systems that operate purely as filtering or blocking mechanisms. By offering explanations and safer alternatives, the system empowers users to make informed decisions rather than restricting their actions.

F. Limitations and Future Directions

Despite its strengths, the system exhibits limitations in handling ambiguous and context-dependent language. Challenges such as over-triggering in safe scenarios and inconsistent detection of informal address expressions indicate the need for more advanced contextual understanding.

Future work may explore improved ambiguity resolution using larger language models, enhanced contextual embeddings, and more diverse training datasets. Additionally, refining risk calibration and improving rewrite consistency are important directions for enhancing overall system reliability.

G. Implications for Human-Layer Cybersecurity

The findings of this study reinforce the importance of shifting from traditional system-level security toward human-centered approaches. As social media continues to be a primary vector for information exposure and exploitation, proactive tools that assist users at the point of content creation become increasingly valuable.

The Guardian framework demonstrates that integrating multiple detection capabilities into a real-time, user-facing system is both feasible and effective, offering a promising direction for future research and deployment in human-layer cybersecurity.

IX. CONCLUSION

This paper presented *The Guardian*, a modular real-time framework designed to address human-layer cybersecurity risks on social media platforms. The system integrates multiple detection modules, including misinformation, hate speech, scam/phishing detection, and PII with contextual OpSec risk inference, within a unified architecture.

The proposed framework demonstrates that combining diverse detection capabilities into a single, user-facing system is both feasible and effective. System-level evaluation shows that the approach performs well in identifying explicit and contextual risks, particularly in PII and OpSec-related scenarios. The results also highlight the system's ability to provide meaningful explanations and actionable feedback, supporting users in making informed decisions before sharing content.

At the same time, the evaluation reveals limitations in handling ambiguous and context-dependent language, as well as a tendency toward conservative predictions in safe scenarios. These findings reflect the inherent challenges of human-centered cybersecurity and indicate areas for further improvement.

Future work will focus on enhancing contextual understanding, improving risk calibration, and refining rewrite generation for more consistent risk reduction. Additionally, expanding the system with more advanced language models and broader datasets may further improve performance across diverse real-world scenarios.

Overall, this work contributes toward the development of proactive, user-centric cybersecurity solutions that operate at the human layer, offering a promising direction for improving online safety in modern digital environments.

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